

Other Classes + metadata before we use Amp2

Note, still assumes stock culture pre culture growth steps are offline
? stock
→ preculture
? Amp2
atm
sample added to plate offline

✓ = compiled elsewhere, just display in LIMS, does not need captured by ADB again
⊗ = Doesn't need captured by a human in LIMS

Stock Culture
input →
⊗ -is-a: Activity
⊗ -activity-id
-instrument-id
? are they ID'd?
just name?
? should ADB treat inst. as entities?
-volume-added
-name
-description
⊗ -type
⊗ -input
-output
? is there output? I think not

Media
input →
⊗ -entity-id
-name
-description
-type
-media-type
-volume-ml
-media-recipe
-media-formulation
-commercial-media-catalog
-serialization-method
-ph-adjustment
-ph-target
-exposure-sensitivity
-media-additions
-storage-temperature

Liquid Handler Prep

Liquid Handler Prep

? is Media + sample added sequentially?

? Entities in LIMS + ADB for each plate

Plate

-is-a: Entity
-name
-description
-type
⊗ -entity-id
-number-of-wells
-well-volume
-well-shape
-lot-number
? do we need to track exact purchase batch of plates? if yes, entities will need updated & this is hard to manage
-pipette-tip-size
-pipette-tip-conductivity
-pipette-tip-filter
-media
-sample
-plate-type
-well-map

controlled by plate + liquid handler

	1	2	3	4
A	media vol sample vol			
B				
C				

? Create well map as a separate protocol?

input

Experimental Culture

⊗ -is-a: Culture Growth
⊗ -activity-id
-name
-description
⊗ -type
-input
-output
-treatment
? treatments are likely to be a well by well distinction,
-plate-id
? media plate?
combine the two?
-incubation-time-hours
? or should this live on the output entity?
-temperature-celsius
-agitation-speed
✓ -oxygen-status
? move to earlier step?
✓ -container-type
? provided by input?
✓ -substrates
-start-time
-growth-time

Analysis DB

well-A1
well-A2
well-A3
well-A4
well-B1
-media-volume
-strain-volume
-well-id
? sample id?
-treatment
? granularity?

? constrained by liquid handler & plate details

OD Measurement

⊗ -is-a: Activity
-name
-description
⊗ -type
-input
-output
-od-method
-wavelength
-read-times
? list, exact times or time after Exp Cult start-time?

output

Data File

? a lot of metadata about this file is in the file or the LIMS Workflow.
Good check, ADB just needs to do a check

input

Data Processing

? this is the conversion script, ? wrote to convert the weird time into actual time

output

Processed Data File